

ECE448 Spring 2025

Lesson 33

AI5



UNITED STATES
**AIR FORCE
ACADEMY**

NOMINAL SCHEDULE

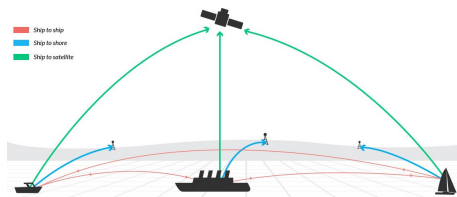
Lesson	Topics	Notes
22	RF Reverse Engineering	Final Project Proposal due
23	RF Reverse Engineering	
24	RF Reverse Engineering	
25	Flex day	RF RE Project Due
26	ADS-B	
27	Project Work Day	Background Research due
28	ADS-B	
29	ADS-B	
30	AIS	
31	AIS	Design Description due
32	WBFM Radio	ADS-B/AIS Project
33	HD Radio	
34	HD Radio	
35	Project Work Day	Prototype Description due
36	QAM	
37	QAM /OFDM	
38	QAM /OFDM	
39	Project Day	
40	Final Project Presentations	Final Project due

LESSON 33 LEARNING OBJECTIVES

- 1.1 Articulate the purpose and implementation of the Automated Identification System (AIS)
- 1.2 Understand the encoding and modulation schemes of AIS
- 1.3 Use existing tools to receive and decode AIS messages

AUTOMATIC INFORMATION SYSTEM (AIS)

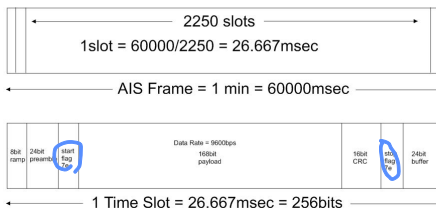
- Allows ships to automatically identify themselves
- Broadcasts various classes of messages on VHF (161.975MHz & 162.025MHz)
- Self Organizing Time Division Multiple Access (SOTDMA)
 - Each slot is 26.66 ms
 - Two channels, 2250 slots per frame, 1 frame per minute
 - 9600 baud
- Non-Return to Zero Inverted (NRZI) encoding, GMSK Modulation



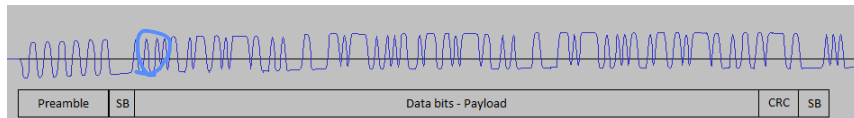
Source:
<https://shipping.nato.int/nsc/operations/news/2021/ais-automatic-identification-system-overview>

AIS MESSAGE STRUCTURE

- 8-bit ramp up
- 24-bit preamble (01010...)
- 8-bit start flag (0x7e)
- 168-bit payload
- 16-bit CRC



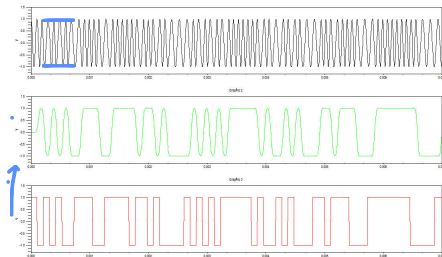
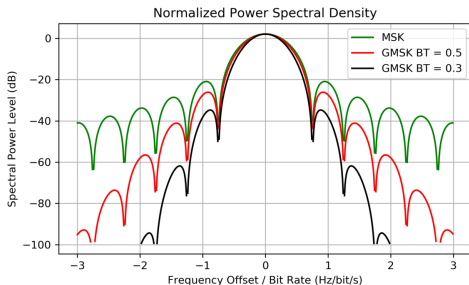
Source: <https://jeremyclark.ca/wp/nav/ais-modulator-scicos-simulation/>



Source: https://en.wikipedia.org/wiki/Automatic_identification_system

GMSK MODULATION

- Gaussian Minimum Shift Keying (GMSK)
- Type of FSK
- Reduces power in harmonics
- Constant envelope



EXAMPLE DATA

NMEA 0183 governs format, data sent as printable ASCII:

```
!AIVDM,1,1,,A,14eG;o@034o8sd<L9i:a;WF>062D,0*7D
```

!AIVDM	NMEA message type, other NMEA device messages are restricted
1	Number of sentences (some messages need more than one, maximum generally is 9)
1	Sentence number (1 unless it is a multi-sentence message)
	Blank is the sequential message ID (for multi-sentence messages)
A	AIS channel (A or B), for dual channel transponders it must match the channel used
14eG; . . .	Encoded AIS data, using AIS-ASCII6
0*	End of data, number of unused bits at end of encoded data (0-5)
7D	NMEA checksum (NMEA 0183 Standard CRC16)

EXAMPLE DATA: AIS PAYLOAD

The data payload is:

!AIVDM,1,1,,A,14eG;o@034o8sd<L9i:a;WF>062D,0*7D

Which needs to be interpreted as 6-bit ASCII and then parsed bitwise according to its message type.

Field	Bits
Message ID	6
Repeat Indicator	2
MMSI User ID	30
⋮	⋮

TOOLS

- [gr-ais](#)
- [gr-satellites](#)
- SDR Angel
- SCEPTRE

CRC

- Cyclic Redundancy Check
- CRC16 = 16 bits
- Checksum for data
- Easily generated using CRC library in Python
- Can be used for error correction

```
from crc import Calculator, Crc8
data = bytes((0, 1, 2, 3, 4, 5))
calculator = Calculator(Crc8.CCITT)
checksum = calculator.checksum(data) #expected = 0xBC
print(f"CRC8-CCITT checksum: {checksum:02x}")
```

```
# Verify checksum
is_valid = calculator.verify(data, checksum)
print(f"Is checksum valid: {is_valid}")
```

ACTIVITIES

- Generate NMEA sentence using pyais
- Check out the Sigmf file from the Noir CTF challenge – use whatever tool you like
- Parse payload data
- Try to send AIS data to someone
 - Could use [AIVDM generator](#), NRZI encode, GMSK mod
 - Could also use [AIS Simulator](#)
 - Another approach?